LOCALIZING A MORPHISM OF RINGED SITES: THE CANONICAL (F')^\SHARP AND THE COMMUTATIVE SQUARE OF RINGED TOPOI

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span contains the author's phrase 'everything else follows from' with a precise citation of Sites, Lemma sites-lemma-localize-morphism; kept verbatim and recorded here as the author's own reduction step, not as an omission.

Editorial note. The counted span cites one Sites lemma, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Localization square for a morphism of topoi. Stacks Project, Sites, Lemma sites-lemma-localize-morphism. Supplies the commutative square of topoi and the functor behaviour that the ringed statement refines; also underlies the identification (f')^{-1}j_V^{-1} = j_U^{-1}f^{-1} used in the composite.
- j^{-1} = j^* on sheaves of modules for a localization morphism. Stacks Project, Modules on Sites, discussion of localization morphisms. Used in the author's closing remark to pass from the first functor equality to the second.

Source reference. Original reference 03CF: sources/stacks-chow/tags.txt, source lines 4382–4382. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(f, f^\sharp) : (\mathcal{C}, \mathcal{O}) \longrightarrow (\mathcal{D}, \mathcal{O}')$ be a morphism of ringed sites where f is given by the continuous functor $u : \mathcal{D} \to \mathcal{C}$. Let V be an object of $\mathcal{D}$ and set $U = u(V)$. Then there is a canonical map of sheaves of rings $(f')^\sharp$ such that the diagram of Sites, Lemma 03CF is turned into a commutative diagram of ringed topoi*

$$\begin{array}{ccc} (Sh(\mathcal{C}/U), \mathcal{O}_U) & \xrightarrow[(j_U, j_U^\sharp)]{} & (Sh(\mathcal{C}), \mathcal{O}) \\ {\scriptstyle (f', (f')^\sharp)}\downarrow & & \downarrow{\scriptstyle (f, f^\sharp)} \\ (Sh(\mathcal{D}/V), \mathcal{O}'_V) & \xrightarrow{(j_V, j_V^\sharp)} & (Sh(\mathcal{D}), \mathcal{O}'). \end{array}$$

Moreover, in this situation we have $f'_* j_U^{-1} = j_V^{-1} f_*$ and $f'_* j_U^* = j_V^* f_*$.

Proof. Just take $(f')^\sharp$ to be

$$(f')^{-1}\mathcal{O}'_V = (f')^{-1} j_V^{-1} \mathcal{O}' = j_U^{-1} f^{-1} \mathcal{O}' \xrightarrow{j_U^{-1} f^\sharp} j_U^{-1} \mathcal{O} = \mathcal{O}_U$$

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and everything else follows from Sites, Lemma 03CF. (Note that $j^{-1} = j^*$ on sheaves of modules if j is a localization morphism, hence the first equality of functors implies the second.) $\square$